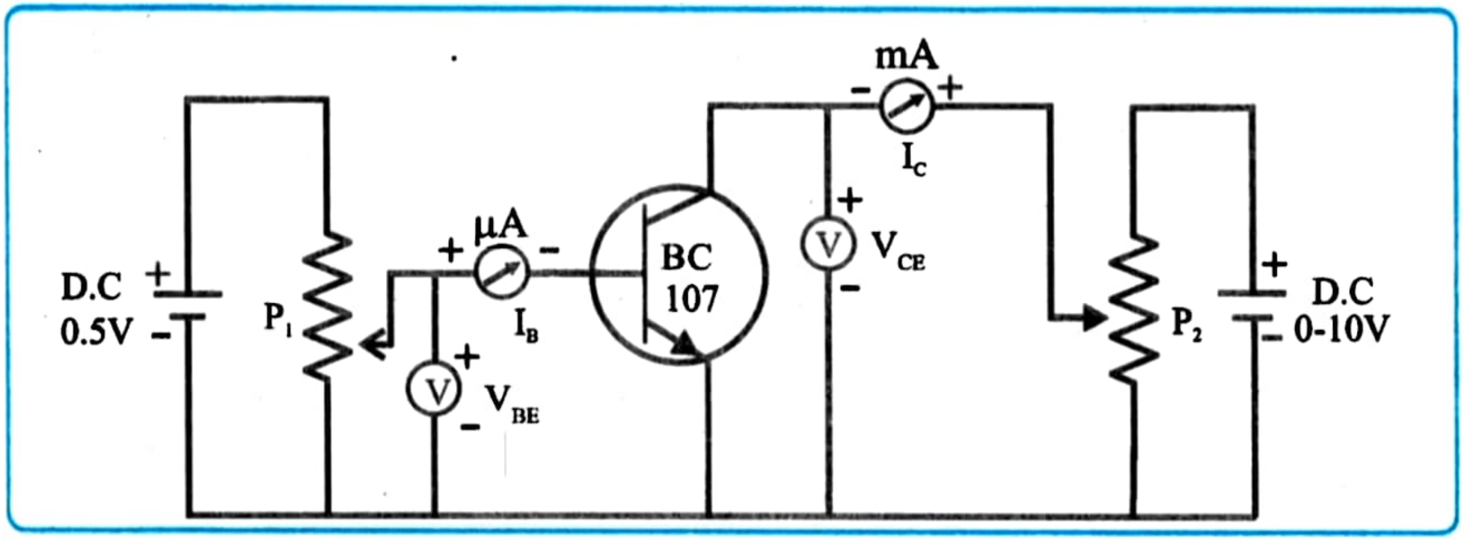
EXPERIMENT NO. 15
CHARACTERISTICS OF TRANSISTER

Aim : To study input and output characteristics of a common emitter NPN transistor.

Apparatus : Power supply, micro ammeter, mili-ammeter, voltmeter.

Circuit Diagram:



Components :

Observation Table : 1) Input Characteristics

Obs. No.	$V_{CE} = 5V$ at $I_B = 0$	
	I_B (μA)	V_{BE} (V)
1	0	0.2
2	0	0.3
3	0	0.4
4	0	0.5
5	15	0.6
6	140	0.7
7	10	1.75
8	20	4
9	30	5.5
10	40	7.5

2) Output Characteristics

Obs. No.	$I_B = 10\ \mu A$		$I_B = 20\ \mu A$		$I_B = 30\ \mu A$	
	I_C (mA)	V_{CE} (V)	I_C (mA)	V_{CE} (V)	I_C (mA)	V_{CE} (V)
1	1.5	0.1	2.75	0.1	3.5	0.1
2	3.75	0.2	6.25	0.2	8.75	0.2
3	3.75	0.3	7.5	0.3	10.5	0.3
4	3.75	0.4	7.5	0.4	10.75	0.4
5	3.75	0.5	7.5	0.5	10.75	0.5
6	3.75	0.6	7.5	0.6	10.75	0.6
7	3.75	0.7	7.5	0.7	10.75	0.7
8	3.75	0.8	7.5	0.8	10.75	0.8
9	3.75	0.9	7.5	0.9	10.75	0.9
10	3.75	1	7.5	1	10.75	1

Calculations :

From Output Characteristic graph.

1) For $I_B = 20\ \mu A$

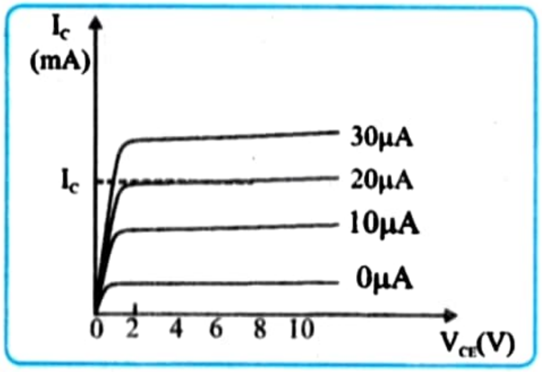
I_C = ...13.... mA

2) Current amplification factor $\beta = I_C / I_B$ = 6.50 .

Graph :

1) Plot a graph of I_B against V_{BE} for input characteristic (Graph 1)

2) Plot a graph of I_C against V_{CE} for output characteristic (Graph 2)



Conclusion :

The input characteristic is similar to the forward bias characteristic of PN junction diode.

Result : Current amplification factor $\beta = I_C / I_B$ = 6.50

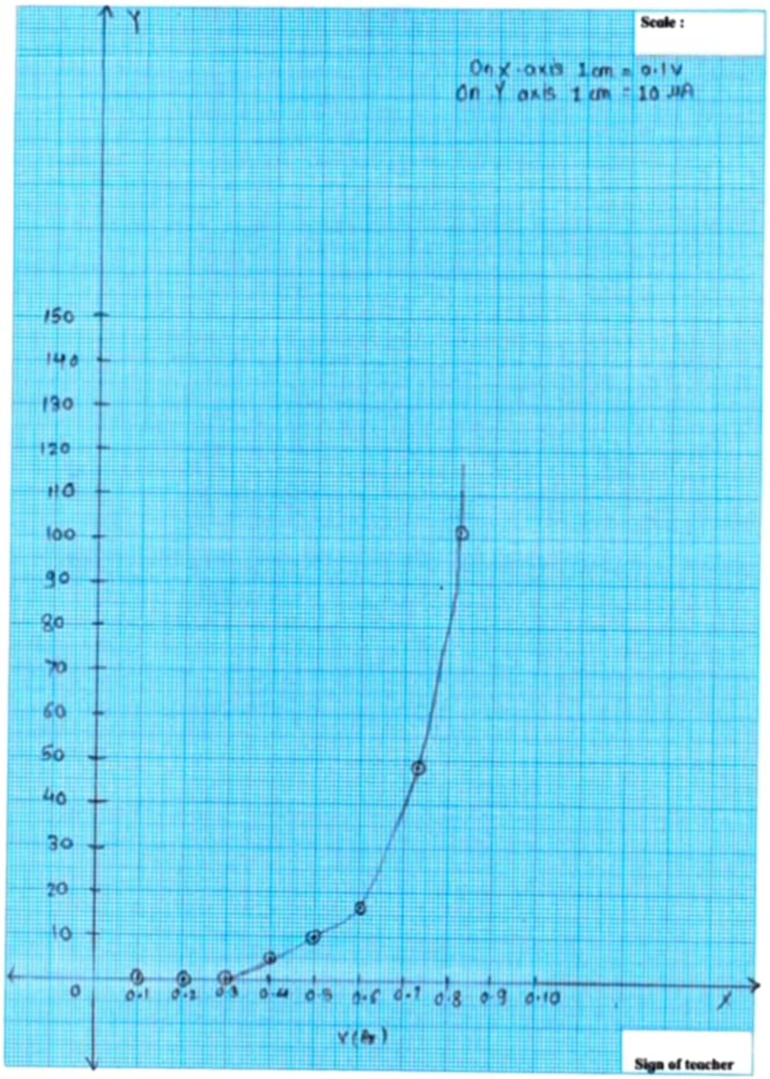
Questions

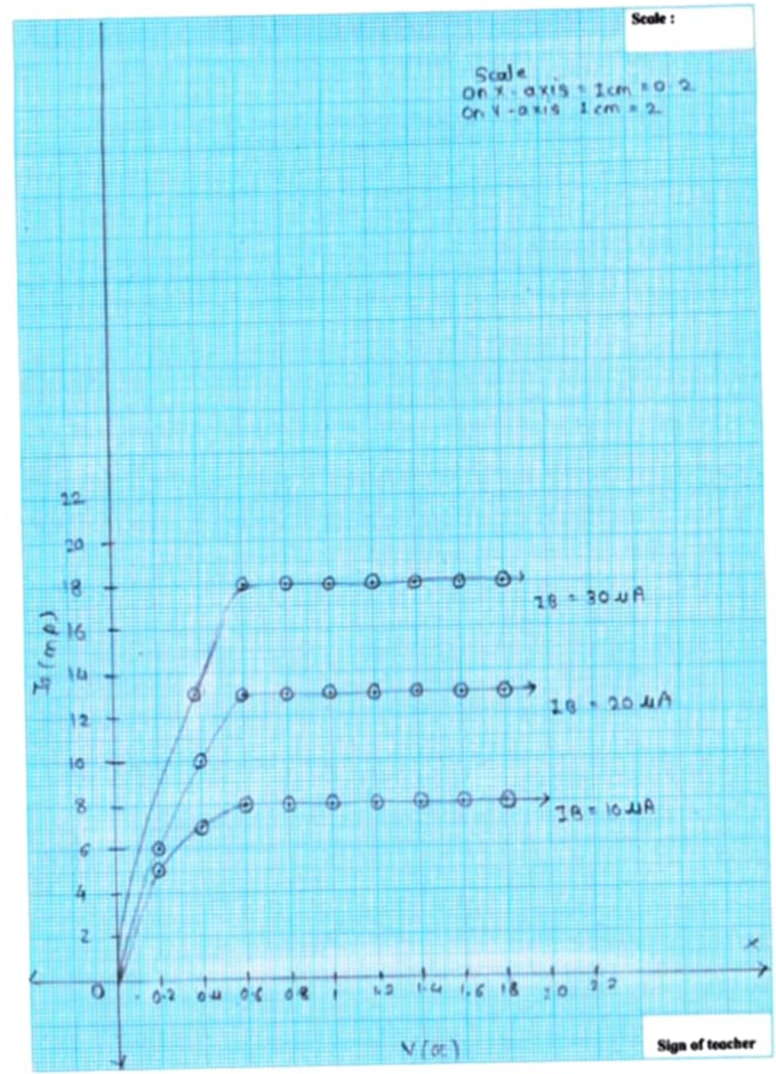
1. What do you mean by common emitter configuration ?

The configuration in which the emitter is connected between the collector and base is known as a common emitter configuration.

2. How biasing is done for a transistor?

Transistor biasing can beachieved either by using a single feed back resistor or by using a simple voltage divider network to provide the required biasing voltage.


Scale :
On X-axis 1cm = 0.1V
On Y axis 1 cm = 10 µA
Y
150
140
130
120
110
100
90
80
70
60
50
40
30
20
10
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
0.10
X
Sign of teacher


Scale :
Scale
On X-axis 1cm = 0.2
On Y-axis 1cm = 2
22
20
18
16
14
12
10
8
6
4
2
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2.0
2.2
X
IB = 30 µA
IB = 20 µA
IB = 10 µA
V(CE)
Sign of teacher